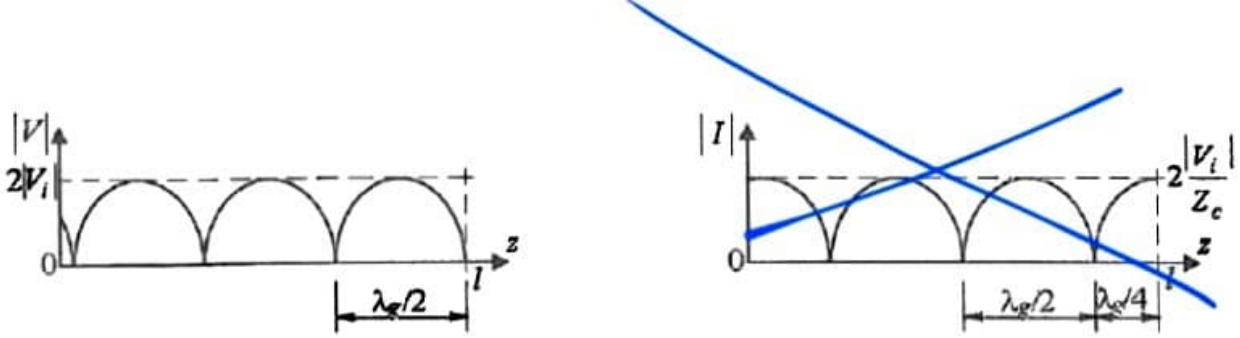


Fig. 1.4: Distribuția tensiunii și curentului pe linie, pentru $Z_s = 0$

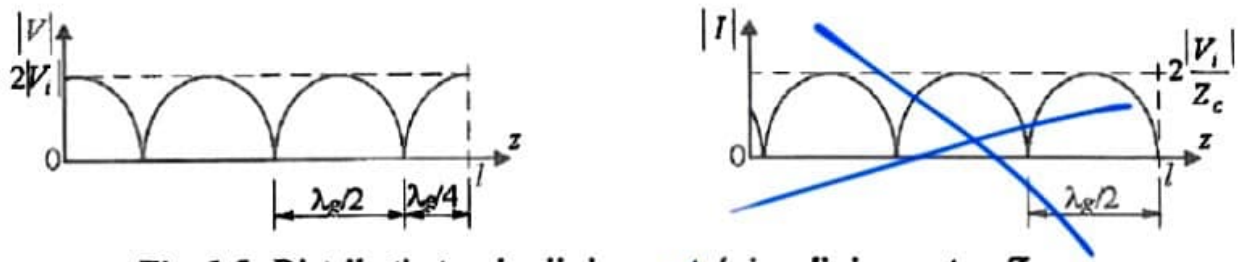


Fig. 1.5: Distribuția tensiunii și curentului pe linie, pentru $Z_s = \infty$

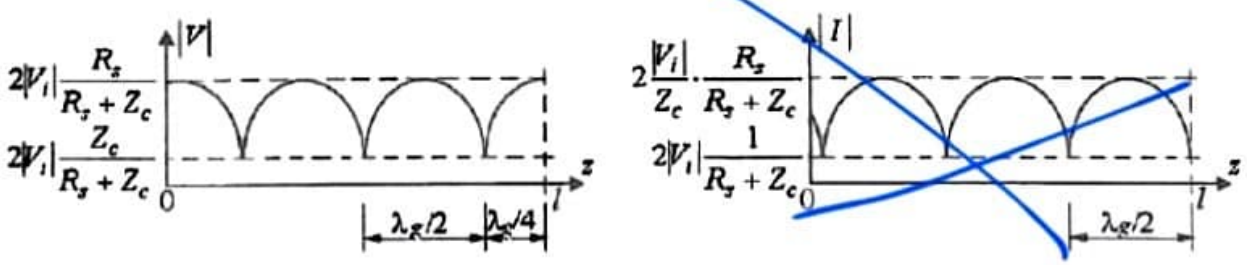


Fig. 1.6: Distribuția tensiunii și curentului pe linie, pentru $Z_s = R_s > Z_c$

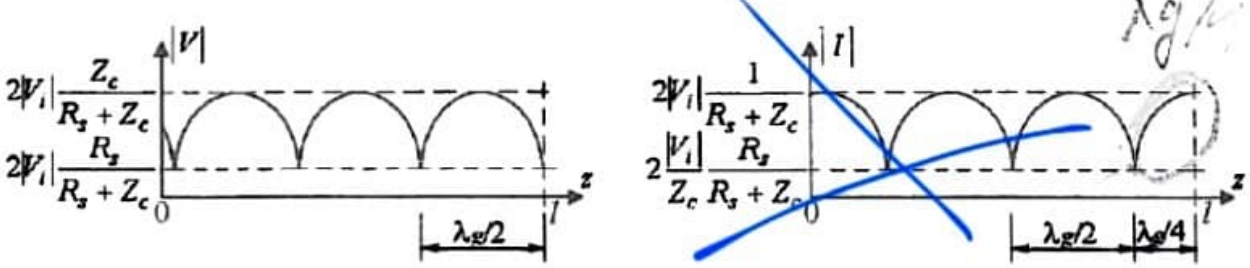


Fig. 1.7: Distribuția tensiunii și curentului pe linie, pentru $Z_s = R_s < Z_c$

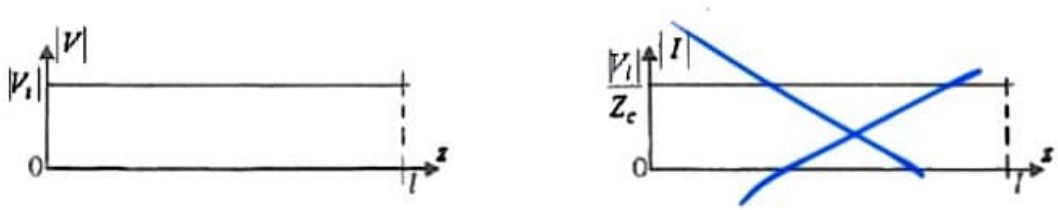


Fig. 1.8: Distribuția tensiunii și curentului pe linie, pentru $Z_s = Z_c$

FORMULE

$$\sigma = \frac{1+|\Gamma_s|}{1-|\Gamma_s|}$$ → raportul de undă staționară

$$|\Gamma_s| = \frac{\sigma - 1}{\sigma + 1}$$ → modulul coef. de reflexie

$$\Gamma_s = \frac{Z_s - Z_c}{Z_s + Z_c} \qquad \Gamma_{in} = \frac{Z_{in} - Z_c}{Z_{in} + Z_c}$$

$$Z_s = Z_c \cdot \frac{1+\Gamma_s}{1-\Gamma_s} \qquad \Gamma_{in} = \frac{V_r}{V_i}$$

$Z_s \to 0 \Rightarrow \Gamma_s = -1 \Rightarrow$ linie term. în sc.

$Z_s \to \infty \Rightarrow \Gamma_s = 0 \Rightarrow$ linie term. în gol

$Z_s \to Z_c \Rightarrow \Gamma_s = 0 \Rightarrow$ linie term. adaptat

$$\Gamma_s = |\Gamma_s| \cdot e^{j\varphi_{\Gamma_s}}$$

$$|\Gamma_s| = \frac{\sqrt{a^2+b^2}}{c}$$

$$\varphi_{\Gamma_s} = -\operatorname{arctg}\frac{b}{a}$$

λ_g = lung. de undă pe linie

$\theta = \beta l$ $\qquad$ θ = lung. electrică

$\beta = \frac{2\pi}{\lambda_g}$ $\qquad$ β = const. de defazare

- Dist. dintre 1 minim și un maxim: $\frac{\lambda_g}{4}$
- Dist. dintre 2 minime / maxime de tens: $\frac{\lambda_g}{2}$
- Distanța la primul maxim de tens.

$$\Delta z = \frac{\varphi_{\Gamma_s} \cdot \lambda_g}{4\pi}$$

$$\lambda_g = \frac{\lambda_0}{\sqrt{\varepsilon_r}} = \frac{c_0}{f_0} \cdot \frac{1}{\sqrt{\varepsilon_r}}$$

- Distanța la primul minim de tens.

$$d_{min} = \frac{\lambda_g}{4} + \Delta z$$

$$Z_{in} = Z_c \cdot \frac{Z_s + jZ_c \cdot tg(\beta l)}{Z_c + jZ_s \cdot tg(\beta l)}$$

$$Z_s \to \infty \Rightarrow Z_{in} = -j \cdot \frac{Z_c}{tg(\beta l)}$$

$$|V|_{max} = |V_i| \cdot (1 + |\Gamma_s|) = |V_i| + |V_i| \cdot \Gamma_s = |V_i| + |V_r|$$

$$|V|_{min} = |V_i| \cdot (1 - |\Gamma_s|) = |V_i| - |V_r|$$

$\beta l = \frac{\pi}{2} \Rightarrow Z_{in} = \frac{Z_c^2}{Z_s}$

$\beta l = \pi \Rightarrow Z_{in} = Z_s$

$Z_s = 0 \Rightarrow Z_{in} = j Z_c \, tg(\beta l)$

$Z_s = \infty \Rightarrow Z_{in} = -j Z_c \, ctg(\beta l)$

$Z_s = Z_c \Rightarrow Z_{in} = Z_c$

$P_{i,in}$ = puterea complexă incidentă
$P_{r,in}$ = puterea complexă reflectată
$\Rightarrow$ la intrare

$P_{i,in} = \frac{|V_i|^2}{2 Z_c}$

$P_{r,in} = \frac{|V_r|^2}{2 Z_c}$

P_{in} = puterea absorbită la intrare

$P_{in} = \frac{|V_{in}|^2}{2 \cdot |Z_{in}|^2} \cdot Re(Z_{in})$

$P_{in} = P_{i,in} - P_{r,in}$

P_s = puterea absorbită de sarcină

$P_s = P_{i,s} - P_{r,s} = P_{i,s}\left(1 - \frac{P_{r,s}}{P_{i,s}}\right) = P_{i,s}\left(1 - |\Gamma_s|^2\right)$

$\frac{P_{r,s}}{P_{i,s}} = |\Gamma_s|^2$

$\eta = \frac{P_s}{P_{in}}$ $\rightarrow$ randament

RL = return loss

$RL = -10 \lg |\Gamma_s|^2 = -20 \lg |\Gamma_s|$

$a_m [dB] = 10 \lg P_{i,s} - 10 \lg P_s$

PARȚIAL 2023

(1) pierderi neglijabile $\Rightarrow \alpha = 0$

$Z_c = 50\,\Omega$

$\varepsilon_r = 4$

2 max de tens. consec.: $d_1 = 3\,cm$

$d_2 = 6\,cm$

$|\Gamma_s| = 0{,}2$

a) $\lambda_g = ?$

b) $f = ?$

c) VSWR $= ?$

d) grafic

e) $Z_s = ?$

a) $d_2 - d_1 = \frac{\lambda_g}{2} \Rightarrow \lambda_g = 2(d_2 - d_1) = 2(6-3) = 6\,cm$

b) $\lambda_g = \frac{\lambda_0}{\sqrt{\varepsilon_r}} = \frac{c_0}{f\sqrt{\varepsilon_r}} \Rightarrow f = \frac{c_0}{\lambda_g\sqrt{\varepsilon_r}} = \frac{3\cdot 10^8}{6\cdot 10^{-2}\cdot 2} = \frac{1}{4}\cdot 10^{10} = 0{,}25\cdot 10^{10}\,Hz$

$= 2{,}5\,GHz$

c) VSWR $= \sigma$

$\sigma = \frac{1+|\Gamma_s|}{1-|\Gamma_s|} = \frac{1+0{,}2}{1-0{,}2} = \frac{1{,}2}{0{,}8} = 1{,}5$

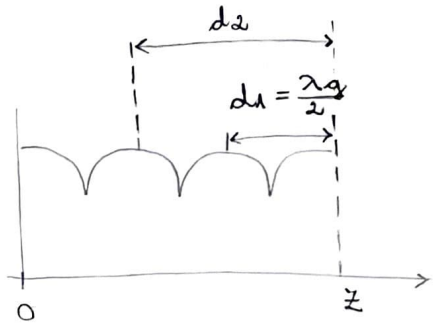
e) dist. la primul max de tens: $\Delta z = 0 \Rightarrow \varphi_{\Gamma_s} = 0 \Rightarrow \Gamma_s = |\Gamma_s|\cdot \underbrace{e^0}_{1} = 0{,}2$

$d_1 = \Delta z = \frac{\varphi_{\Gamma_s}\cdot\lambda_g}{4\pi} \Rightarrow \varphi_{\Gamma_s} = \frac{d_1\cdot 4\pi}{\lambda_g} = \frac{3\cdot 4\pi}{6} = 2\pi$

$\Gamma_s = |\Gamma_s|\cdot e^{j\varphi_{\Gamma_s}} = 0{,}2\cdot e^{j\cdot 2\pi} = 0{,}2\cdot(\underbrace{\cos 2\pi}_{1} + j\cdot\underbrace{\sin 2\pi}_{0}) = 0{,}2$

$Z_s = Z_c\cdot\frac{1+\Gamma_s}{1-\Gamma_s} = 50\cdot\frac{1+0{,}2}{1-0{,}2} = 50\cdot 1{,}5 = 75\,\Omega$

d)



② $Z_c = 50\,\Omega$

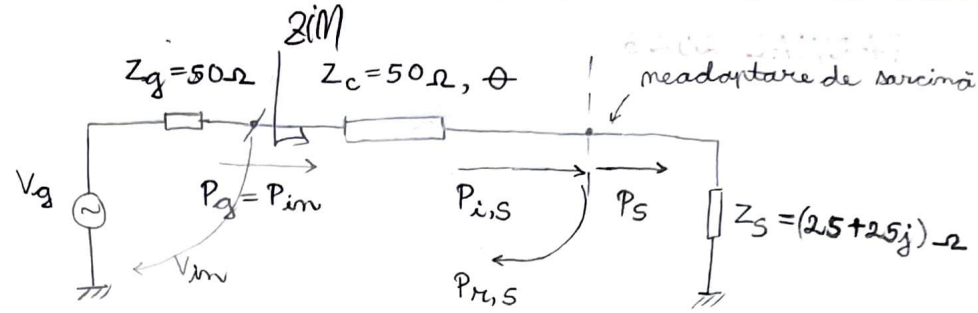
$Z_s = (25 + 25j)\,\Omega$

$l = \frac{\lambda_g}{4}$

$P_{r,s} = 0{,}5\,W$

pierderi neglijabile $\Rightarrow \alpha = 0$

$Z_g = Z_c = 50\,\Omega$



a) $|\Gamma_s| = ?$

b) $P_{i,s} = ?$ $\quad P_s = ?$

c) $P_{i,in} = ?$ $\quad P_{r,in} = ?$

d) $V_g = ?$

a) $\Gamma_s = \frac{Z_s - Z_c}{Z_s + Z_c} = \frac{25+25j-50}{25+25j+50} = \frac{-25+25j}{75+25j} = \frac{\overset{3-j}{-1+j}}{3+j} = \frac{-3+3j+j+1}{9+1} = \frac{-2+4j}{10} = \frac{-1+2j}{5}$

$|\Gamma_s| = \sqrt{\frac{1}{25} + \frac{4}{25}} = \frac{\sqrt{5}}{5}$

b) $P_s = P_{i,s} - P_{r,s} = P_{i,s}\left(1 - \frac{P_{r,s}}{P_{i,s}}\right) = P_{i,s}\left(1 - |\Gamma_s|^2\right)$

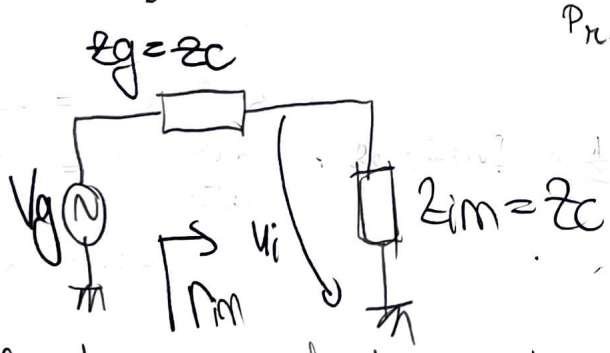
$\frac{P_{r,s}}{P_{i,s}} = |\Gamma_s|^2 \Rightarrow P_{i,s} = \frac{P_{r,s}}{|\Gamma_s|^2} = \frac{0{,}5}{\frac{5}{25}} = 0{,}5 \cdot 5 = 2{,}5\,W$

$P_s = 2{,}5\left(1 - \frac{5}{25}\right) = 2{,}5 \cdot \frac{20}{25} = 2\,W$

c) $\left.\begin{array}{l} \cdot \text{ linia nu are pierderi} \\ \cdot\ Z_c = Z_g \end{array}\right\} \Rightarrow P_{i,in} = P_{in} = 2{,}5\,W$

$P_{r,in} = 0\,W$

d)



$V_i = \frac{Z_c}{Z_c + Z_c} \cdot V_g \Rightarrow V_i = \frac{V_g}{2}$

Când am adaptare de impedanță $\Gamma_{in} = 0$. $\Rightarrow$

$\Rightarrow P_{in} = P_{i,in}(1 - \underbrace{|\Gamma_{in}|^2}_{=0}) \Rightarrow P_{in} = P_{i,in} = 2{,}5$

$P_{i,in} = \frac{|V_i|^2}{2Z_c} = \frac{\left|\frac{V_g}{2}\right|^2}{2Z_c} = \frac{|V_g|^2}{8Z_c} \Rightarrow |V_g| = \sqrt{8Z_c \cdot P_{i,in}}$

$= \sqrt{8 \cdot 2{,}5 \cdot 50} = \sqrt{1000} = 31{,}6$

Aplicația 6

Se dă o linie de transmisiune terminată pe Z_S, pt. care:

$\sigma = 2$

$d = 22$ cm - dist. până la un minim de tensiune (nu neap. primul)

$\lambda_g = 30$ cm

$Z_c = 75\,\Omega$

$Z_S = ?$

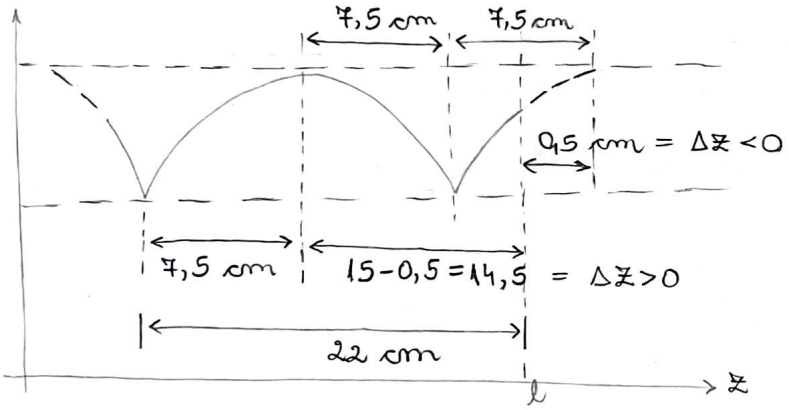
$$\sigma = \frac{1+|\Gamma_S|}{1-|\Gamma_S|} \Leftrightarrow |\Gamma_S| = \frac{\sigma-1}{\sigma+1} = \frac{2-1}{2+1} = \frac{1}{3} = 0{,}33$$

$$\frac{\lambda_g}{4} = \frac{30}{4} = 7{,}5 \text{ cm} < d = 22 \text{ cm}$$

$$\frac{\lambda_g}{2} = \frac{30}{2} = 15 \text{ cm} < d = 22 \text{ cm}$$

$\Rightarrow$ primul minim de tens. este la:

$22 - 15 = 7$ cm $= d'$



$$\Delta z = \frac{\varphi_{\Gamma_S} \cdot \lambda_g}{4\pi} \Rightarrow \varphi_{\Gamma_S} = \frac{\Delta z \cdot 4\pi}{\lambda_g}$$

Pt. $\Delta z < 0$

$$d' = \frac{\lambda_g}{4} - \Delta z \Rightarrow \Delta z = \frac{\lambda_g}{4} - d' = 7{,}5 - 7 = 0{,}5 \text{ cm}$$

$$\varphi_{\Gamma_S} = -\frac{0{,}5 \cdot 4\pi}{30} \cong -0{,}21 \text{ rad}$$

$$\Gamma_S = \frac{Z_S - Z_c}{Z_S + Z_c} \Rightarrow Z_S = Z_c \cdot \frac{1+\Gamma_S}{1-\Gamma_S}$$

$$\Gamma_S = 0{,}33 \cdot e^{-j \cdot 0{,}21}$$

$$Z_S = 75 \cdot \frac{1 + 0{,}33 \cdot \cos(0{,}21) - j \cdot 0{,}33 \cdot \sin(0{,}21)}{1 - [0{,}33 \cdot \cos(0{,}21) - j \cdot 0{,}33 \cdot \sin(0{,}21)]} \cong (143 - 22j)\,\Omega$$

Aplicația 5

Se dă o linie de transmisiune având $Z_c = 50\,\Omega$, terminată pe $Z_S = (20 - 53j)\,\Omega$.

Frecv. de generator este $f_0 = 1$ GHz.

Dielectricul liniei este aerul: $\varepsilon_r = 1$ iar $\alpha = 0$.

$\sigma = ?$

d = dist. până la primul minim de tens.

$$\Gamma_S = \frac{Z_S - Z_c}{Z_S + Z_c} = \frac{20 - 53j - 50}{20 - 53j + 50} = \frac{-30 - 53j}{70 - 53j} = \dots = \frac{709 - 5300j}{7709}$$

$$|\Gamma_S| \cong \frac{5347}{7709} \cong 0{,}69$$

$$\varphi_{\Gamma_S} = \operatorname{arctg} \frac{5300}{709} \simeq -1{,}43 \text{ rad}$$

$$\sigma = \frac{1 + |\Gamma_S|}{1 - |\Gamma_S|} = \frac{1 + 0{,}69}{1 - 0{,}69} = \frac{1{,}69}{0{,}31} = 5{,}45$$

$$\Delta z = \frac{\varphi_{\Gamma_S} \cdot \lambda_g}{4\pi}$$ (dist. până la primul maxim de tens)

$\varphi_{\Gamma_S} < 0 \Rightarrow \Delta z < 0 \Rightarrow$ maxim de tens. la $z > l$

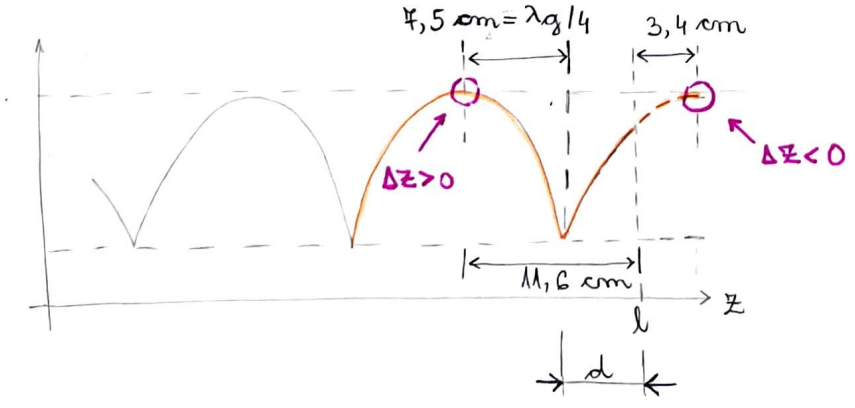
$\varphi_{\Gamma_S} > 0 \Rightarrow \Delta z > 0 \Rightarrow$ maxim de tens. la $z < l$

$$\lambda_g = \frac{\lambda_0}{\sqrt{\varepsilon_{eff}}} = \frac{\lambda_0}{\sqrt{\varepsilon_r}} = \lambda_0 = \frac{c_0}{f} = \frac{3 \cdot 10^8}{10^9} = 0{,}3 \text{ m} = 30 \text{ cm}$$

mediu omogen $\quad$ $\varepsilon_r = 1$

$$\Delta z = \frac{\varphi_{\Gamma_S} \cdot \lambda_g}{4\pi} = \frac{-1{,}43 \cdot 0{,}3}{4 \cdot 3{,}14} \cong -0{,}034 \text{ m} = -3{,}4 \text{ cm}$$

$$d = \frac{\lambda_g}{4} - \Delta z = 7{,}5 - 3{,}4 = 4{,}1 \text{ cm}$$



Pt. $\Delta z > 0 \Rightarrow \varphi_{r_s} = 2\pi - 1{,}43 = 4{,}85$ rad

$$\Delta z = \frac{\varphi_{r_s} \cdot \lambda_g}{4\pi} = \frac{4{,}85 \cdot 0{,}3}{4\pi} \simeq 0{,}116 \text{ m} = 11{,}6 \text{ cm}$$

$$\left(= \frac{\lambda_g}{2} - 3{,}4 \text{ cm}\right)$$

$$d = \Delta z - \frac{\lambda_g}{4} = 11{,}6 - 7{,}5 = 4{,}1 \text{ cm}$$

la fel ca mai sus

PARȚIAL NR. 1

① $\varepsilon_r = 4$

$Z_c = 50\,\Omega$

$Z_s = (25 + 25j)\,\Omega$

$f = 5\,GHz$

---

a) $|\Gamma_s| = ?$ $\varphi_{\Gamma s} = ?$

b) $\Delta z = ?$

c) primul minim de tens.

d) $l = \frac{\lambda_g}{2} \Rightarrow \Gamma_s = ?$

e) $\sigma = ?$

a) $\Gamma_s = \frac{Z_s - Z_c}{Z_s + Z_c} = \frac{25+25j-50}{25+25j+50} = \frac{-25+25j}{75+25j} = \dots = \frac{-2500+2500j}{6250}$

$|\Gamma_s| = \sqrt{0{,}16+0{,}16} = \sqrt{0{,}32} = 0{,}56$

$\varphi_{\Gamma s} = \operatorname{arctg}\frac{b}{a} = \operatorname{arctg}(-1) = -\frac{\pi}{4}$

$\Gamma_s = |\Gamma_s| \cdot e^{j\varphi_{\Gamma s}} = 0{,}56 \cdot e^{-j\frac{\pi}{4}}$

b) $\Delta z = \frac{\varphi_{\Gamma s} \cdot \lambda_g}{4\pi} = \frac{-\frac{\pi}{4} \cdot 0{,}3}{4\pi} = -\frac{0{,}3}{16} = -1{,}87\ cm$

c) $d = \frac{\lambda_g}{4} - 1{,}87 = 7{,}5 - 1{,}87 = 5{,}63\ cm$

d) $l = \frac{\lambda_g}{2} \Rightarrow \theta = 180^\circ \Rightarrow |\Gamma_s| = 0$

e) $\sigma = \frac{1+|0{,}56|}{1-|0{,}56|} = \frac{1{,}56}{0{,}34} = 4{,}58$

③ $\varepsilon_r = 1$

$Z_c = 75\,\Omega$

$Z_s = 75 - 75j\ \Omega$

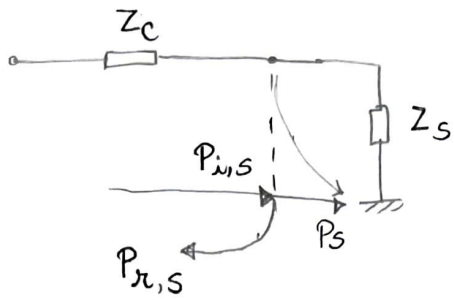
$P_{i,s} = 5W$

a) $P_{r,s} = ?$

b) $P_s = ?$

c) $V_i = ?$

d) $V_r = ?$



$$\Gamma_s = \frac{Z_s - Z_c}{Z_s + Z_c} = \frac{75 - 75j - 75}{75 - 75j + 75} = \dots = \frac{1}{\sqrt{5}}$$

b) $|\Gamma_s|^2 = \frac{1}{5}$

$$P_s = P_{i,s}\,(1 - |\Gamma_s|^2) = 5\left(1 - \frac{1}{5}\right) = 4W$$

c) $P_{i,s} = \dfrac{|V_i|^2}{2 \cdot Z_c} \Rightarrow |V_i| = \sqrt{2 \cdot Z_c \cdot P_{i,s}} = \sqrt{10 \cdot 75} \cong 27{,}38\,V$

d) $P_{r,s} = \dfrac{|V_r|^2}{2 \cdot Z_c} \Rightarrow |V_r| = \sqrt{2 \cdot Z_c \cdot P_{r,s}} = \sqrt{2 \cdot 75 \cdot 1} = \sqrt{150} \cong 12{,}24\,V$

a) $P_s = P_{i,s} - P_{r,s} \Rightarrow P_{r,s} = P_{i,s} - P_s = 1W$

PARȚIAL NR. 2

① $\varepsilon_r = 1$

$Z_c = 75\,\Omega$

$\sigma = 3$

$d = 1$ cm

$f = 10$ GHz

a) $\lambda_g = ?$

b) $|\Gamma_s| = ?$

c) $\varphi_{\Gamma_s} = ?$

d) $Z_s = ?$

e) $Z_{in} = ?$

$l = \frac{\lambda_g}{4}$

a) $\lambda_g = \frac{\lambda_0}{\sqrt{\varepsilon_r}} = \lambda_0 = \frac{c_0}{f} = \frac{3\cdot 10^8}{10\cdot 10^9} = 0{,}03\ m = 3\ cm$

b) $\sigma = \frac{1+|\Gamma_s|}{1-|\Gamma_s|}$

$|\Gamma_s| = \frac{\sigma - 1}{\sigma + 1} = \frac{2}{4} = \frac{1}{2} = 0{,}5$

c) $\Delta z = \frac{\varphi_{\Gamma_s}\cdot \lambda_g}{4\pi} \Rightarrow \varphi_{\Gamma_s} = \frac{\Delta z \cdot 4\pi}{\lambda_g}$

$d = \frac{\lambda_g}{4} + \Delta z \Rightarrow \Delta z = d - \frac{\lambda_g}{4} = 1 - \frac{3}{4} = \frac{1}{4} = 0{,}25$ cm

$\varphi_{\Gamma_s} = \frac{0{,}25\cdot 4\pi}{3} = \frac{\pi}{3} = 60^\circ = 1{,}04$ rad

d) $Z_s = Z_c \cdot \frac{1+\Gamma_s}{1-\Gamma_s}$

$\Gamma_s = |\Gamma_s|\cdot e^{j\cdot\varphi_{\Gamma_s}} = \frac{1}{2}\left(\cos\frac{\pi}{3} + j\sin\frac{\pi}{3}\right) = \frac{1}{4} + \frac{\sqrt{3}}{4}j$

$Z_s = 75\cdot \frac{\frac{5+\sqrt{3}j}{4}}{\frac{3-\sqrt{3}j}{4}} = 75\cdot \frac{5+\sqrt{3}j}{\cancel{4}}\cdot\frac{\cancel{4}}{3-\sqrt{3}j} = 75\cdot\frac{(15+5\sqrt{3}j + 3\sqrt{3}j - 3)}{12}$

$= 25\cdot\frac{12+8\sqrt{3}j}{4} = 25\cdot(3+2\sqrt{3}j) = 75 + 50\sqrt{3}j$

e) $\beta l = \frac{2\pi}{\lambda_g}\cdot\frac{\lambda_g}{4} = \frac{\pi}{2} = \theta$

$tg\,\theta = 0$

$Z_{in} = Z_s = 75 + 50\sqrt{3}j$

③ $\varepsilon_r = 4$

$Z_c = 50\,\Omega$

$Z_s = 25 - 25j$

$Z_g = 50\,\Omega$

$V_g = 10V$

a) $P_{i,s} = ?$

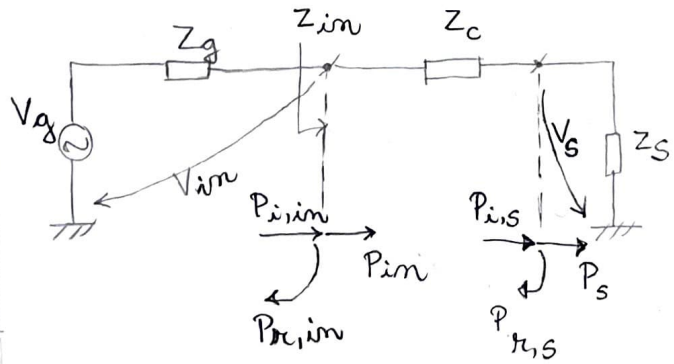
b) $P_{r,s} = ?$

c) $P_s = ?$

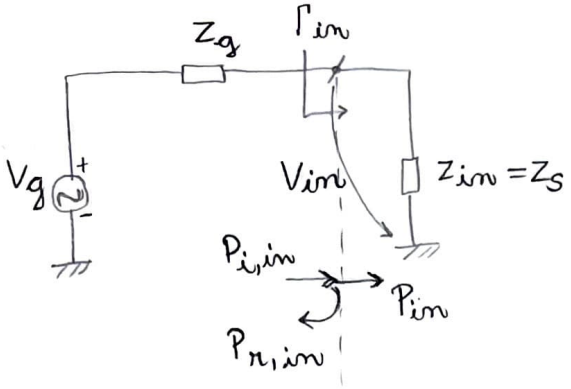
d) $\sigma = ?$

e) $P_{in} = ?$

$l = \frac{\lambda_g}{2}$



$efp \Rightarrow \begin{cases} P_{in} = P_s \\ P_{i,in} = P_{i,s} \\ P_{r,in} = P_{r,s} \end{cases}$



$$P_{in} = \frac{|V_{in}|^2}{2\cdot|Z_{in}|^2}\cdot Re\{Z_{in}\}$$

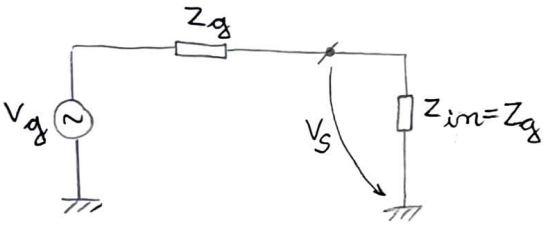
$$V_{in} = \frac{Z_{in}}{Z_{in}+Z_g}\cdot V_g = \frac{25-25j}{25-25j+50}\cdot 10 = \frac{25-25j}{75-25j}\cdot 10 = \frac{\overset{3+j}{\cancel{1-j}}}{3-j}\cdot 10 = \frac{\cancel{10}(1-j)(3+j)}{\cancel{10}}$$

$$=(4-2j)\ V$$

$$P_{in} = \frac{(\sqrt{16+4})^2}{2\cdot(\sqrt{625+625})^2}\cdot 25 = \frac{20}{2\cdot 2\cdot \cancel{625}_{25}}\cdot \cancel{25} = \frac{\cancel{20}^{5}}{\cancel{4}\cdot 25} = \frac{1}{5} = 0{,}2\,W = P_S$$

$$P_{in} = P_{i,in} - P_{r,in} = P_{i,in}\,(1-|\Gamma_{in}|^2)$$

Dacă $\Gamma_{in}=0 \Rightarrow P_{in} = P_{i,in}$



$$V_S = \frac{Z_g}{Z_g+Z_g}\cdot V_g = \frac{1}{2}V_g = 5V$$

$$P_{i,in} = \frac{V_S^2}{2\cdot Z_g} = \frac{25}{100} = \frac{1}{4} = 0{,}25\,W$$

$$P_{in} = P_{i,in} - P_{r,in} \Rightarrow P_{r,in} = P_{i,in} - P_{in} = 0{,}25 - 0{,}2 = 0{,}05\,W$$

$$\Gamma_{in} = \frac{Z_{in}-Z_g}{Z_{in}+Z_g} = \frac{25-25j-50}{25-25j+50} = \frac{-25-25j}{75-25j} = \frac{-1-j}{3-j} = \dots = \frac{-1-2j}{5}$$

$$|\Gamma_{in}| = \frac{\sqrt{5}}{5}$$

d) $$\Gamma_S = \frac{Z_S-Z_C}{Z_S+Z_C} = \frac{25-25j-50}{25-25j+50} = \dots = \frac{-1-j}{3-j} = \dots = \frac{-1-2j}{5}$$

$$|\Gamma_S| = \frac{\sqrt{5}}{5}$$

$$\sigma = \frac{1+|\Gamma_S|}{1-|\Gamma_S|} = \frac{1+\frac{\sqrt{5}}{5}}{1-\frac{\sqrt{5}}{5}} = \frac{\overset{5+\sqrt{5}}{\cancel{5+\sqrt{5}}}}{5}\cdot\frac{5}{5-\sqrt{5}} = \frac{(5+\sqrt{5})^2}{25-5} = \frac{30+10\sqrt{5}}{20} = \frac{3+\sqrt{5}}{2}$$

e) $$\theta = \beta l = \frac{2\pi}{\cancel{\lambda_g}}\cdot\frac{\cancel{\lambda_g}}{2} \Rightarrow \boxed{\theta = \pi}$$

$$\Rightarrow Z_{in} = Z_S = 25-25j$$

$$\Rightarrow P_{in} = P_S = 0{,}2\,W$$

Aplicația 1

$Z_c = 100\ \Omega$

$P_s = 10\ W$

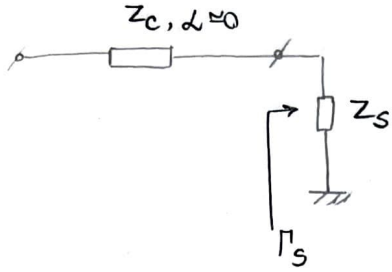
$Z_s = (50 + 150j)\ \Omega$

$\alpha \simeq 0$

$|V|_{max} = ?$

$|V|_{min} = ?$

$\sigma = ?$



$$\Gamma_s = \frac{Z_s - Z_c}{Z_s + Z_c} = \frac{50 + 150j - 100}{50 + 150j + 100} = \frac{-50 + 150j}{150 + 150j} = \frac{-1 + 3j}{3 + 3j}$$

$$\Rightarrow |\Gamma_s| = \frac{\sqrt{1+9}}{\sqrt{9+9}} = \frac{\sqrt{10}}{\sqrt{18}} = \frac{\sqrt{5}}{3}$$

$$|\Gamma_s|^2 = \frac{5}{9}$$

$$P_s = P_{i,s} - P_{r,s} = P_{i,s}\left(1 - \frac{P_{r,s}}{P_{i,s}}\right) = P_{i,s}\left(1 - |\Gamma_s|^2\right)$$

$$P_{i,s} = \frac{P_s}{1 - |\Gamma_s|^2} = \frac{10}{1 - \frac{5}{9}} = \frac{90}{4} = 22,5\ W$$

$$P_{r,s} = P_{i,s} - P_s = 22,5 - 10 = 12,5\ W$$

$$P_{i,s} = \frac{|V_i|^2}{2 Z_c} \Rightarrow |V_i| = \sqrt{2 \cdot Z_c \cdot P_{i,s}} = \sqrt{2 \cdot 100 \cdot 22,5} \cong 67\ V$$

$$P_{r,s} = \frac{|V_r|^2}{2 Z_c} \Rightarrow |V_r| = \sqrt{2 \cdot Z_c \cdot P_{r,s}} = \sqrt{2 \cdot 100 \cdot 12,5} \cong 50\ V$$

$$|V|_{max} = |V_i| + |V_r| = 67 + 50 = 117\ V$$

$$|V|_{min} = |V_i| - |V_r| = 67 - 50 = 17\ V$$

$$\sigma = \frac{|V|_{max}}{|V|_{min}} = \frac{117}{17} \simeq 6,9$$

sau: $\sigma = \dfrac{1 + |\Gamma_s|}{1 - |\Gamma_s|} = \dfrac{1 + \sqrt{\frac{5}{9}}}{1 - \sqrt{\frac{5}{9}}} = \dfrac{3 + \sqrt{5}}{3 - \sqrt{5}} \simeq 6,9$

Aplicatia 2

$Z_c = 50\,\Omega$

$\alpha \cdot l \simeq 0$

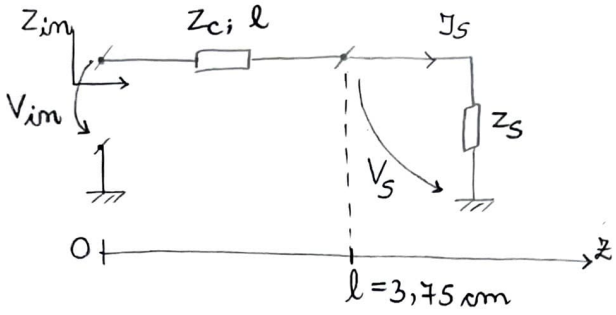
$Z_s = 100\,\Omega$

$f = 1\,GHz$

$\varepsilon_r = 1$

$V_{in} = 10\,V$

$P_s = ?$



$$\lambda_g = \lambda_0 = \frac{c_0}{f} = \frac{3\cdot 10^8}{10^9} = 0{,}3\,m = 30\,cm$$

$$\theta = \beta \cdot l = \frac{2\pi}{\lambda_g}\cdot l = 2\pi \cdot \frac{3{,}75}{30} = 2\pi \cdot \frac{1}{8} = \frac{\pi}{4} \Rightarrow \theta = 45^\circ \quad (tg\,45^\circ = 1)$$

* Metoda 1

$$Z_{in} = Z_c \cdot \frac{Z_s + j\cdot Z_c \cdot tg\,\theta}{Z_c + j\cdot Z_s \cdot tg\,\theta} = 50 \cdot \frac{100+50j}{50+100j} = 50\cdot\frac{2+j}{1+2j} = \overset{10}{\cancel{50}}\cdot\frac{(2+j)(1-2j)}{\underset{1}{\cancel{5}}} =$$

$$= 10\,(2-4j+j+2) = (40-30j)\ \Omega$$

$$P_{in} = P_s = \frac{|V_{in}|^2}{2\cdot|Z_{in}|^2}\cdot Re(Z_{in}) = \frac{100}{2\cdot(1600+900)}\cdot \overset{20}{\cancel{40}} = \frac{20}{25} = \frac{4}{5} = 0{,}8\,W$$

putrea activă în sarcină

* Metoda 2

$$V_{in} = V_s \cdot \cos\theta + j\cdot Z_c \cdot I_s \cdot \sin\theta \quad , \quad I_s = \frac{V_s}{Z_s}$$

$$V_{in} = V_s \cdot \left(\cos\theta + j\cdot Z_c \cdot \frac{1}{Z_s}\cdot \sin\theta\right)$$

$$\Rightarrow V_s = \frac{V_{in}}{\cos\theta + j\cdot\frac{Z_c}{Z_s}\cdot\sin\theta} = \frac{10}{\frac{\sqrt{2}}{2}\cdot\left(1+j\cdot\frac{50}{\underset{2}{\cancel{100}}}\right)} = \frac{4\cdot 10}{\sqrt{2}\,(2+j)}\ V$$

$$P_s = \frac{|V_s|^2}{2 Z_s} = \frac{1600}{2\cdot 2\cdot 5}\cdot\frac{1}{100} = \frac{16}{4\cdot 5} = \frac{4}{5} = 0{,}8\ W$$

Z_s real

Aplicația 3

$Z_c = 50\Omega$

$\theta = 180°$

$\left(l = \frac{\lambda_g}{2}\right)$

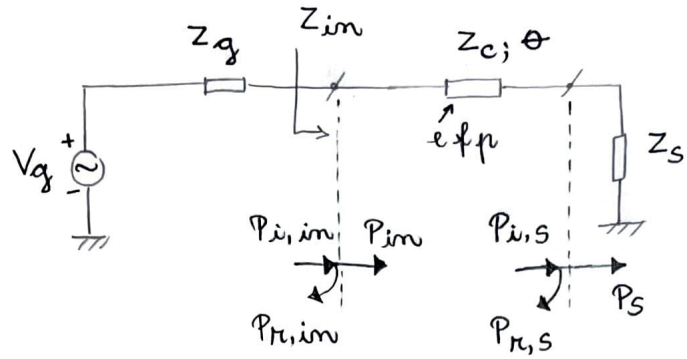
$V_g = 10V$

$Z_g = 50\Omega$

$Z_s = 75\Omega$

$P_s,\ P_{i,s},\ P_{r,s} = ?$

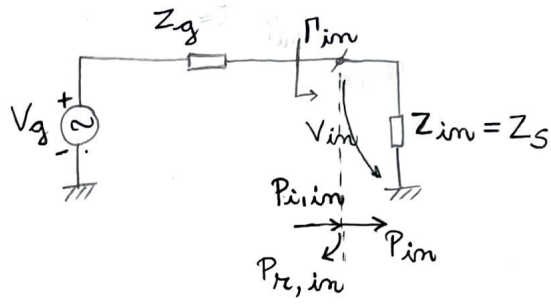
$P_{d,g} = ?$



$\theta = 180°$ @ $f_0 \Rightarrow Z_{in} = Z_s$ @ f_0

$\boxed{Z_{in} = 75\Omega}$ @ f_0

efp: $\begin{cases} P_{in} = P_s \\ P_{i,in} = P_{i,s} \\ P_{r,in} = P_{r,s} \end{cases}$



$$P_{in} = \frac{V_{in}^2}{2 \cdot Z_{in}}$$

$$V_{in} = \frac{Z_{in}}{Z_{in} + Z_g} \cdot V_g = \frac{75}{75+50} \cdot 10 = 6V$$

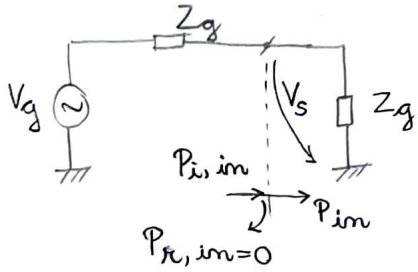
$$P_{in} = \frac{6^2}{2 \cdot 75} = 0{,}24\ W$$

• Vrem să det. și $P_{i,in}$; $P_{r,in}$ apoi verif. $P_{in} = P_{i,in} - P_{r,in}$

$$P_{in} = P_{i,in} \cdot (1 - |\Gamma_{in}|^2)$$

dacă $\Gamma_{in} = 0$ atunci $P_{in} = P_{i,in} = P_{d,g}$

$\Rightarrow P_{i,in}$ se det. din schema:



$$V_s = \frac{Z_g}{Z_g + Z_g} \cdot V_g = \frac{1}{2} \cdot V_g = \frac{10}{2} = 5V$$

$$P_{i,in} = P_{d,g} = \frac{V_s^2}{2 Z_g} = \frac{25}{2 \cdot 50} = 0{,}25W$$

Dar: $\frac{P_{r,in}}{P_{i,in}} = |\Gamma_{in}|^2$ pt schema cu $Z_{in} = 75\Omega$

$$\Gamma_{in} = \frac{Z_{in} - Z_g}{Z_{in} + Z_g} = \frac{75-50}{75+50} = \frac{25}{125} = \frac{1}{5} = 0{,}2$$